More than 67.3188% of the zeros of the Riemann zeta function are simple and lie on the critical line

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Abstract

Let $N(T, 2T)$ count the zeros of the Riemann zeta function with ordinates in $(T, 2T]$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line in that interval. We prove

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq 0.673188803503\ldots > \frac{673188}{10^6}.$$

The proof combines the analytic framework of Claude [1] (Theorem D and its Lean artifact), the stability refinement of the rank–trace argument introduced in [3], and the re-optimized window, position-weighted seven-point inequality, and sharp block profile of [4]. The single new mathematical ingredient is a strengthened certified constant for the weighted seven-point inequality of [4]: we certify $F \geq 127/25000$ where $1/200$ was certified before, and re-optimize the block length. All finite claims are certified by Arb interval arithmetic in two independent runs; the deduction is reproduced in full.

1 Statement and imported inputs

Throughout, $N(T, 2T)$ counts zeros of ζ with ordinates in $(T, 2T]$ with multiplicity, and $N_0^s(T, 2T)$ counts the simple zeros on the critical line in that range.

Theorem 1.1.

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \;\geq\; \frac{251\, H_{\mathrm{cert}} - \eta\, \frac{3}{1150} \cdot 250}{251 - R} \;=\; 0.67318880350350022\ldots \;>\; \frac{673188}{10^6},$$

where $H_{\mathrm{cert}} = 672457/10^6$, $A = 6223/5000$, $R = 2\sqrt{A} - 1$, *and* $\eta = R/A$.

The proof template is that of [4], which in turn builds on [3] and [1]. We import the following two statements exactly as they are established (and used) there; everything else in this paper is self-contained.

Imported Input 1.2 (Stability counting inequality)**.** *Let* $v : [-\frac{1}{2}, \frac{1}{2}] \to [\frac{3}{4}, 1]$ *be an even trigonometric polynomial (an* admissible window profile*), used in the test family of [1, §2] with the standard boundary ramp. Write*

$$K_v(x) = \int_{-1/2}^{1/2} v(t)\cos(2\pi x t)\, dt, \qquad k_v = \frac{K_v}{K_v(0)}, \qquad H(v) = 2 - \frac{1}{c_1(v)},$$

where $c_1(v) = \left(\int v\right)^2 / \left(\int v^2 + \iint |s - s'| v(s) v(s')\, ds\, ds'\right)$. *Then, with* $S = N_0^s(T, 2T)$ *and* $N = N(T, 2T)$,

$$S \;\geq\; H(v)\, N + \operatorname{tr}\Psi(M) - o(N), \tag{1}$$

where M is the Gram matrix of the retained simple-zero atoms, $\Psi(t) = (t-1)^2$ for $0 \le t \le 2$ and $2t-3$ for $t \ge 2$, and the entries of M satisfy $M_{ij} = k_v(x_i - x_j) + o(1)$ uniformly for normalized ordinate differences in compact sets, the exceptional $o(N)$ zeros being deleted.

Input 1.2 is the stability form of the rank–trace argument: the analytic content (explicit formula, Gabor trace asymptotics for admissible windows, tail estimates, Gram-entry asymptotics) is from [1] and its Lean artifact [2]; the retention of the defect $\operatorname{tr}\Psi(M)$ is the contribution of [3], audited independently in [4] and formalized in Lean in [5]. The deduction below uses (1) only through the block/pinching mechanism of Section 5.

2 The window

We use the window of [4] unchanged:

$$v(s) = \sum_{j=0}^{6} c_j \cos(\omega_j s), \qquad \omega = (\sqrt{2},\, 2\pi,\, 4\pi,\, 6\pi,\, 8\pi,\, 10\pi,\, 12\pi), \tag{2}$$

$$c = \left(10^9,\ 3322500,\ -7609135,\ 1190194,\ -731476,\ -1680572,\ 1141360\right)/10^9.$$

Proposition 2.1 (Certified window bounds)**.** *On $[-\frac{1}{2}, \frac{1}{2}]$ one has $\frac{3}{4} \le v \le 1$ (certified enclosures: $\min v = 0.750213\ldots$, $\max v = 0.995632\ldots$), and*

$$H(v) = 0.67245704141454\ldots \ \ge\ H_{\mathrm{cert}} := \frac{672457}{10^6}.$$

Verification. Arb interval arithmetic over exact rational data; the enclosures above are reproduced on every run of the `fast` module of the verifier (see Section 6). This reproduces the corresponding certificate of [4]. □

3 The sharp block profile

Lemma 3.1. *Let $G \succeq 0$ be Hermitian with $E := 2\sum_{i<j} |G_{ij}|^2$. Then*

$$\operatorname{tr}\Psi(G) \ \ge\ h(E), \qquad h(E) = \begin{cases} E, & 0 \le E \le 1, \\ 2\sqrt{E} - 1, & E \ge 1, \end{cases}$$

and h is sharp.

Proof. For $\lambda \ge 0$ write $x = (\lambda - 1)^2$. Since $\Psi(\lambda) = (\lambda-1)^2 - \big((\lambda-2)_+\big)^2$ and, for $\lambda \ge 1$, $(\lambda - 2)_+ = (\sqrt{x} - 1)_+$ while for $0 \le \lambda < 1$ both $(\lambda-2)_+ = 0$ and $x \le 1$, one has the exact identity $\Psi(\lambda) = g\big((\lambda-1)^2\big)$ with

$$g(x) = \begin{cases} x, & 0 \le x \le 1, \\ 2\sqrt{x} - 1, & x \ge 1. \end{cases}$$

The function g is nonnegative, increasing, concave, and $g(0) = 0$; hence g is subadditive, so for the eigenvalues λ_i of G,

$$\operatorname{tr}\Psi(G) = \sum_i g\big((\lambda_i - 1)^2\big) \ \ge\ g\Big(\sum_i (\lambda_i - 1)^2\Big) = g\big(\|G - I\|_{\mathrm{F}}^2\big) \ \ge\ g(E),$$

using that $\|G - I\|_{\mathrm{F}}^2 = \sum_i (\lambda_i - 1)^2$ and that the off-diagonal part alone contributes E. Sharpness: take $G = I$ except for one eigenvalue $1 + \sqrt{E}$ (with a rank-one off-diagonal realization), giving $\operatorname{tr}\Psi(G) = 2\sqrt{E} - 1$ for $E \ge 1$ and E for $E \le 1$. □

Corollary 3.2 (Chord bound)**.** *For $0 \le E \le A$ one has $h(E) \ge \frac{h(A)}{A} E$; and h is nondecreasing.* $\square$

4 The certified weighted seven-point inequality

Let $w = k_v^2$ for the window (2), let $g_1, \ldots, g_6 \ge 0$ be consecutive normalized gaps with partial sums $y_j = g_1 + \cdots + g_j$ $(y_0 = 0)$, and define

$$F(g_1, \ldots, g_6) = \frac{1}{2300} \sum_{i=1}^{6} g_i + \sum_{0 \le i < j \le 6} a_{ij}\, w(y_j - y_i), \tag{3}$$

with the reflection-symmetric exact rational weights a_{ij} of Table 1. Every span capacity is exactly $\sum_i a_{i,i+r} = 2$ $(r = 1, \ldots, 6)$.

$10^6 a_{ij}$	$j = 1$	2	3	4	5	6
$i = 0$	239252	528172	965879	1000000	1000000	2000000
1		381335	465776	34121	0	1000000
2			379413	12104	34121	1000000
3				379413	465776	965879
4					381335	528172
5						239252

Table 1: The pair weights of [4], used unchanged.

Proposition 4.1 (Certified local inequality)**.** *For all $g_1, \ldots, g_6 \ge 0$,*

$$F(g_1, \ldots, g_6) \;\ge\; \varepsilon_0 := \frac{127}{25000} = 0.00508. \tag{4}$$

Verification. Exhaustive interval subdivision with Arb enclosures of w on a grid of mesh $1/4000$, exactly the decision procedure of [4] with the target raised from $1/200$ to $127/25000$: one-body pruning, per-box range-minimum lower bounds over all 21 pair separations, a certified convex tangent bound, and bisection of the widest coordinate; the run fails loudly at any unresolved terminal cell. Two independent runs certified (4); see Section 6 for statistics and recorded certificates. (The observed minimum of F is ≈ 0.005091029, at a lattice-like configuration; the certified target retains a safety margin.) $\square$

5 Deduction of the main theorem

Set

$$q = 6, \qquad m = 251, \qquad p = \frac{1}{2300}, \qquad B_p = 6p = \frac{3}{1150},$$

$$A = \varepsilon_0\,(m - q) = \frac{127}{25000} \cdot 245 = \frac{6223}{5000} = 1.2446, \qquad R = h(A) = 2\sqrt{A} - 1, \qquad \eta = \frac{R}{A}.$$

Numerically $R = 1.23123284307129\ldots$, $\eta = 0.98925987712621\ldots$.

5.1 Windows to blocks

Let B be a block of m consecutive retained simple zeros with normalized ordinates $y_1 < \cdots < y_m$, Gram matrix G_B, and $E_B = 2\sum_{i<j} |(G_B)_{ij}|^2$. Summing (4) over the $m-q$ consecutive seven-point windows of B: a pair spanning r gaps lies in at most $7-r$ windows and its weight totals at most the span capacity 2; each gap lies in at most 6 windows, contributing at most $6p\sum g_i = B_p\,\mathrm{span}(B)$. By the Gram-entry asymptotics of Input 1.2, $2\sum_{i<j} w(y_j - y_i) \le E_B + o(1)$. Hence

$$E_B + B_p\,\mathrm{span}(B) \;\ge\; A - o(1). \tag{5}$$

5.2 Blocks to defect

We claim

$$\mathrm{tr}\,\Psi(G_B) + \eta\, B_p\,\mathrm{span}(B) \;\ge\; R - o(1). \tag{6}$$

If $E_B \ge A$, then Lemma 3.1 and monotonicity give $\mathrm{tr}\,\Psi(G_B) \ge h(E_B) \ge h(A) = R$. If $E_B < A$, the chord bound (Corollary 3.2) gives $\mathrm{tr}\,\Psi(G_B) \ge \eta E_B$, and (5) yields $\eta E_B + \eta B_p\,\mathrm{span}(B) \ge \eta A - o(1) = R - o(1)$.

5.3 Pinching and averaging

Exactly as in [3, §5] (and [4, §4]): for each of the m offsets, partition the retained simple zeros into consecutive blocks of size m; pinching preserves $\mathrm{tr}\,\Psi$ ($X \mapsto \mathrm{tr}\,\Psi(X)$ is convex and unitarily invariant), so summing (6) over the full blocks of one offset and averaging over the m offsets gives, with $S^\circ = S - o(N)$ retained zeros of total normalized span at most $N + o(N)$, each interior gap lying in a block span for at most $m-1$ of the m offsets,

$$\mathrm{tr}\,\Psi(M) \;\ge\; \frac{R}{m}\,S - \eta\,B_p\,\frac{m-1}{m}\,N - o(N). \tag{7}$$

5.4 Conclusion

Insert (7) and Proposition 2.1 into (1):

$$S \;\ge\; H_{\mathrm{cert}}\,N + \frac{R}{m}S - \eta B_p \frac{m-1}{m} N - o(N),$$

hence

$$\liminf_{T\to\infty} \frac{S}{N} \;\ge\; \frac{m\,H_{\mathrm{cert}} - \eta\,B_p\,(m-1)}{m-R} = \frac{251 \cdot \frac{672457}{10^6} - \eta \cdot \frac{3}{1150} \cdot 250}{251 - R} = 0.6731888035035022379\ldots$$

The final arithmetic (including the enclosure of R and η and the comparison with $673188/10^6$) is itself certified in Arb by the verifier's `fast` module. This proves Theorem 1.1. ☐

5.5 Sanity gates

With $(\varepsilon_0, p, m) = (1/200,\ 1/2300,\ 257)$ the same formula gives $0.673137630699\ldots$, the bound of [4]; with the unit-cap profile ($\eta = 1$, $R = A \le 1$) and the parameters of [3] ($\varepsilon_0 = 19/5000$, $p = 1/3000$, $m = 269$, Montgomery–Taylor window) it gives $0.673008527927\ldots$, and with $\varepsilon_0 \to 0$ it degenerates to Theorem D of [1] ($0.672500703679\ldots$). The verifier additionally reproduces the certificates of both predecessors end-to-end.

6 Reproducibility

All finite claims (Propositions 2.1 and 4.1 and the final arithmetic) are certified by Arb interval arithmetic over exact rational inputs. Source, recorded certificates, and instructions are at

`https://github.com/npip99/zeta-zeros`

(the verifier is that of [4], vendored under MIT with only the design constants changed). Proposition 4.1 was certified in two independent runs of the identical decision procedure: one recorded in this repository (grid 1/4000; statistics in `certificates/`), and one performed independently during the design exchange (1,544,746 nodes, maximum depth 46, eight workers, 419 s). The Lean 4 formalization of the framework ([2, 5]) covers Input 1.2's stability layer, the aggregation scheme, and the kernel-limit asymptotics; certifying the present constants in Lean is mechanical given the β-parametric capstone theorem of [5] and is left to a future artifact revision.

Credit

This note is a minimal increment on [4] (whose window, weights, profile, and verifier are used unchanged), which in turn extends [3] over the framework of [1]. The improvement was found and cross-verified in a Claude–GPT-5.6 collaboration on August 11, 2026.

References

[1] Claude, *More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, preprint, August 10, 2026, available online.

[2] Claude, *zeta-23-lean: a Lean 4 formalization*, `https://github.com/anthropics/zeta-23-lean`, 2026.

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[4] T. Haugland et al., *zeta-simple-zeros-673137: a 67.3137630699% lower bound for simple zeros of the Riemann zeta function*, research artifact, August 11, 2026, `https://github.com/trmdy/zeta-simple-zeros-673137`.

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[7] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in: Analytic Number Theory (St. Louis, 1972), Proc. Sympos. Pure Math. **24**, Amer. Math. Soc., 1973, 181–193.